

Exact finite certificates for the Lih–Wang endpoint and cycle-family results

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Internal supplement; not peer reviewed.

This supplement states exactly what each finite computation proves. It is part of the proof package for the accompanying manuscript. No computation below is a sample of matrices or parameter values standing in for an unproved continuous claim.

1 Endpoint gate in orders 2 through 31

For each $2 \leq n \leq 31$ and $1 \leq j \leq n$, the endpoint checker constructs the two rational Bernstein quantities $s_{n,j}$ and $d_{n,j}$ in (2.10)–(2.11) of the manuscript and proves them strictly positive. The zeroth Bernstein layer is an identity. Thus all 495 nontrivial layers of the universal one-collision envelope are closed exactly.

Run from the repository root:

```
.\.venv\Scripts\python.exe -X utf8 loops\LIH-WANG-DIRECT-SUM-GATE-0001\verification\endpoint_threshold_guard.py
```

Frozen script SHA-256:

92134EF380BB170B429C1DA581B8B120525A94004D8818636831183CDA5030B5

Expected certificate digest:

9b857ec7f67ca6319d3cab31c2b3f8fd88690766b51898ffbb577c155a9107e

2 All-transposition shared-center stars

For $3 \leq n \leq 6$, and every permitted star degree $2 \leq k < n$, the certificate consists of exact Bernstein coefficients on the complete interval $0 \leq a \leq 1/k$. The verifier reconstructs the matrix from its row description, recomputes the rook coefficients, checks the certificate, and then changes the claimed certificate to confirm that the destructive control is rejected. It does not import the discovery formula.

```
.\.venv\Scripts\python.exe -X utf8 loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\verify_star_certificate.py --certificate loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\results\star_low_degree_certificates.json --output tmp\lih-wang-star-verification.json
```

Frozen SHA-256 values:

- verifier: 6B62DABBC0E1DC8C88AF635199DC52D72A14A0CF69C981F2D281FBD227710164;
- certificate: 1DA27D8D7B4BA33CC2F73B5FBC876958FF806509AD2357CD967394D79D12FFA5;
- independent reconstruction receipt:
754D34F07532FD1F97FB00EB5A971762B85B04060626F4F5B7F2FFE758DEB4D5.

Order $n = 7$ is not included in these continuous certificates. It is closed by the analytic permanent-threshold gate used in the manuscript; the same verifier checks that threshold arithmetic only.

3 Shared-center orders 8 through 31

Two exact scalar certificates close the continuous parameter ranges left after the analytic endpoint and collision reductions.

For $8 \leq n \leq 15$, the coupled small-mean checker verifies 92 Bernstein layers together with the four coupling gates and a control obtained by dropping the deterministic unit term:

```
.\venv\Scripts\python.exe -X utf8 loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\coupled_small_mean_guard.py
```

Script SHA-256: 9f9c15ec33ac51ae7a725240e5c2d030efff8c75fe497f4992b1e0cc3e600da4.

Expected digest: fcec1ecd6696659d57cce642eaa5ca73b2736fd92c0bb8dbb0bcb60920c49a1e.

For $16 \leq n \leq 31$, the Poisson/Bernoulli boundary checker verifies 376 Bernstein layers after proving the scalar split and monotonicity conditions:

```
.\venv\Scripts\python.exe -X utf8 loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\poisson_boundary_window_guard.py
```

Script SHA-256: fa9b0396fe24441d9aa4a7c5a7871e31999aa79bb615db1ffb2d323cce8a1eba.

Expected digest: 7a8f89627a7afb141eefc19f9d1d60ba4eb49bbafcdcd757ebbb1bd109c82a3.

4 Non-star shared-center structures in orders 3 through 7

After short-branch symmetrization, the complete non-star range has 32 support structures. The discovery packet subdivides their full stick-breaking parameter domains into 318 exact rational Bernstein boxes. The independent checker reconstructs each original matrix and every required coefficient without calling the discovery generator.

```
.\venv\Scripts\python.exe -X utf8 loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\run_small_shape_independent_window.py --packet loops\LIH-WANG-NONCUBICAL-FACES-0001\verification\results\small_shape_window_s32_v1.json --output tmp\lih-wang-small-shape-independent.json
```

Frozen SHA-256 values:

- discovery packet: E50A431535547C10B6A08FECA98F4287A7A03731150E0E3F938B8563E574EE61;
- independent checker: 8D9F3C147BDB354EE103A9BEA04BD824F902BA39224CBE6C167C6097B5C4F84B;
- window runner: 36E693C2AC1BE509363EE9942287576BE6ADBC33F677626C3645F5B440BBBF77;
- frozen independent receipt: FEEA45B5E2275A75664D51E210BC8360C00C3252203241B2423477276F4231F6.

The word “complete” here refers only to the enumerated shared-center support structures in orders 3 through 7, after the stated symmetrization.

5 Scope and trust boundary

The scripts prove the finite rational statements listed above. They do not prove:

- the universal one-collision envelope;
- the midpoint-equivalence theorem for $n \geq 32$;

- the entropy lower bound or one-hot decorrelation lemma;
- the independent-cycle product theorem;
- the analytic estimates for $n \geq 32$;
- any novelty or priority claim.

Those obligations are paid in the manuscript. The full frozen dependency list and source hashes are recorded in `loops/LIH-WANG-PUBLICATION-PREP-0001/A1-写作前审计包.md`.